Sequence-to-Sequence Models: Intro to Attention

Naeemullah Khan
naeemullah.khan@kaust.edu.sa

جامعة الملك عبدالله
للعلوم والتقنية
King Abdullah University of
Science and Technology

KAUST Academy
King Abdullah University of Science and Technology

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Motivation

Why Do We Need Seq2Seq and Attention?

- Many real-world problems require transforming one sequence to another:
  - Translation: `"Bonjour"` → `"Hello"`
  - Dialogue systems: Question → Response
  - Speech: Audio → Text

Standard RNNs struggle with input/output sequences of different lengths and long-term dependencies.

Seq2Seq models + attention solve this with a powerful encoder-decoder framework.

Learning Outcomes

By the end of this session, you should be able to:

- Explain the Seq2Seq architecture and encoder-decoder framework
- Understand the bottleneck problem in fixed-size representations
- Describe the motivation for and core idea behind attention mechanisms
- Appreciate how attention improves performance in NLP tasks
- Recognize future directions in attention-based modeling

Seq2Seq Architecture

Seq2Seq Architecture

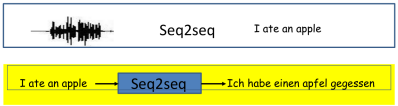
Key Idea: Map input sequence $\rightarrow$ intermediate vector $\rightarrow$ output sequence.

- **Encoder RNN:** Processes input sequence and compresses it into a **fixed-length vector** (context).
- **Decoder RNN:** Generates output sequence from the context vector.

Applications:

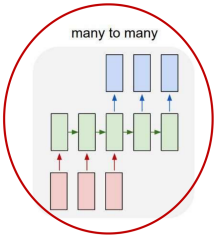
- Machine translation
- Summarization
- Dialogue systems
- Speech recognition

Seq2Seq Architecture



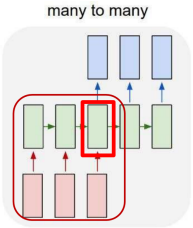
- Sequence goes in, sequence comes out
- No notion of "time synchrony" between input and output
  - May even not even maintain order of symbols
    - E.g. "I ate an apple" → "Ich habe einen apfel gegessen"
  - Or even seem related to the input
    - E.g. "My screen is blank" → "Please check if your computer is plugged in."

Modeling the problem



Delayed sequence to sequence

Modeling the problem



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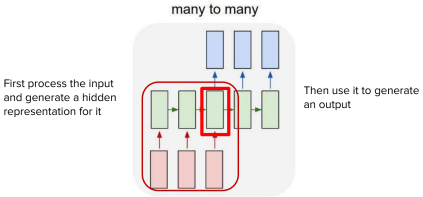

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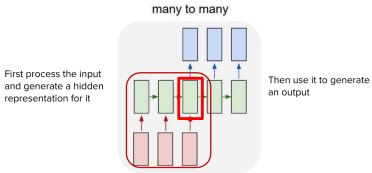
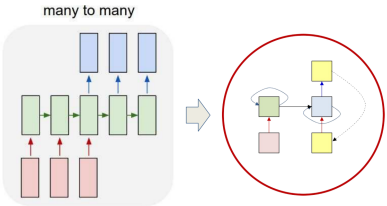


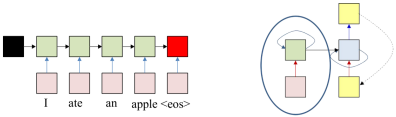
Problem: Each word that is output depends only on current hidden state, and not on previous outputs

Modeling the problem



- *Delayed* Sequence to sequence
  - Delayed *self-referencing* sequence-to-sequence

The “simple” translation model



- The input sequence feeds into a recurrent structure
- The input sequence is terminated by an explicit <eos> symbol
  - The hidden activation at the <eos> “stores” all information about the sentence

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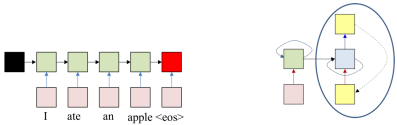

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- Subsequently a second RNN uses the hidden activation as initial state, and <sos> as initial symbol, to produce a sequence of outputs
  - The output at each time becomes the input at the next time
  - Output production continues until an <eos> is produced

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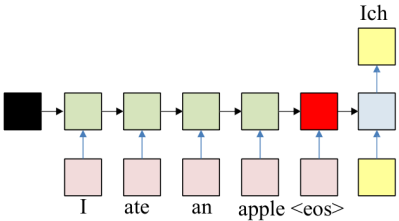

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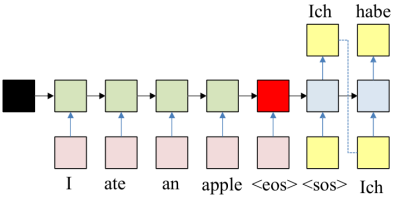

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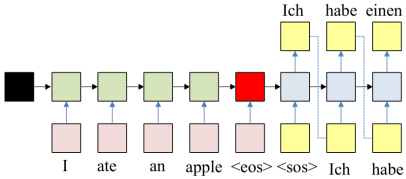

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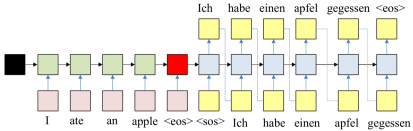

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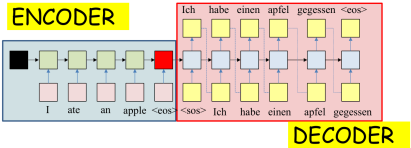


- The recurrent structure that extracts the hidden representation from the input sequence is the **encoder**
- The recurrent structure that utilizes this representation to produce the output sequence is the **decoder**

The Decoding Problem

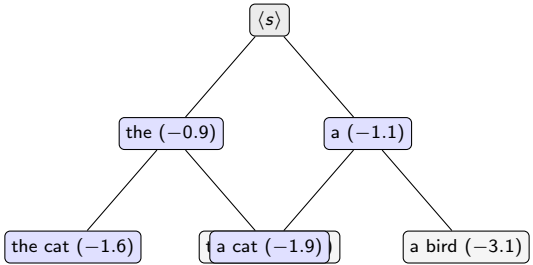
- At each step the decoder outputs a probability distribution over the vocabulary. Decoding is the search for the output sequence with the highest overall probability:

$$\hat{y} = \arg\max_{y} \prod_{t=1}^{T} P(y_t \mid y_{<t}, x)$$

- Searching over all possible sequences is intractable: $|V|^T$ candidates for vocabulary size $|V|$ and length T.
- **Greedy decoding:** take the single most probable token at every step.
- Greedy is cheap but short-sighted: a locally best word can commit the decoder to a worse overall sentence, and there is no way back.

Beam Search

- Keep the k best partial hypotheses (the **beam**) at each step instead of just one.
- At every step, extend each hypothesis with every vocabulary token, score all candidates by cumulative log-probability, and keep the top k.
- $k = 1$ recovers greedy decoding; machine translation typically uses beams of 4 to 10.



Beam of $k = 2$: at each step all extensions are scored and only the two best partial sequences (blue) survive.

Beam Width Trade-offs and Length Normalization

- **Wider beams** cost more compute and give diminishing returns; very wide beams can even hurt quality by surfacing short, generic outputs.
- **Length bias:** summing log-probabilities makes every added token lower the score, so beam search favors short outputs.
- **Length normalization** divides the score by T^{α} (with $\alpha \approx 0.6$ in Wu et al., 2016) so hypotheses of different lengths compete fairly.
- Beam search suits tasks with a well-defined target (translation, summarization). For open-ended generation it tends to produce repetitive, generic text, which is why sampling-based decoding is used there instead.

The Bottleneck Problem

The Bottleneck Problem

Fixed-length context vector = information bottleneck

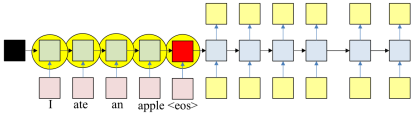
- Encoder must **compress entire input sequence** into a single vector
- Longer or more complex inputs $\rightarrow$ information loss
- Decoder relies solely on that vector to produce outputs

Leads to poor performance on long sentences or tasks requiring high context awareness

A problem with this framework

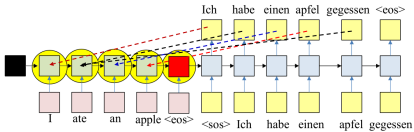
- All the information about the input sequence is embedded into a single vector.
  - The "hidden" node layer at the end of the input sequence.
  - This one node is "overloaded" with information, particularly if the input is long.

A problem with this framework



- In reality: All hidden values carry information.
  - Some of which may be diluted by the time we get to the final state of the encoder.

A problem with this framework



- Every output is related to the input directly.
- Not sufficient to have the encoder hidden state to only the initial state of the decoder.
- Misses the direct relation of the outputs to the inputs.

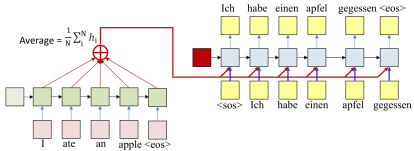
Realization: We Need More Context!

Realization: We Need More Context!

Decoder should have access to all encoder states, not just the final one.

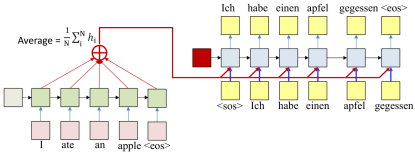
- This inspired the development of the **attention mechanism**.
- Instead of passing only the final state, allow the decoder to *"look back"* at **all input positions**.

Using all input hidden states



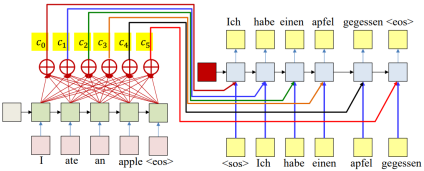
- Simple solution: Compute the average of all encoder hidden states.
- Input this average to every stage of the decoder.
- The initial decoder hidden state is now separate from the encoder and may be a learnable parameter.

Using all input hidden states



- Problem: The average applies the same weight to every input.
- It supplies the same average to every output word.
- In practice, different outputs may be related to different inputs.
  - E.g., “Ich” is most related to “I”, and “habe” and “gegessen” are both most related to “ate”.

Using all input hidden states

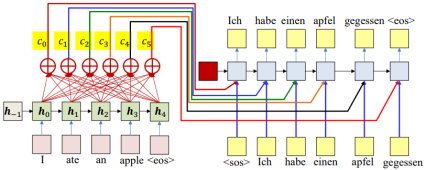


Solution: Use a *different* weighted average for each output word

- The weighted average provided for the kth output word is:

$$c_t = \sum_i w_i(t) h_i$$

Using all input hidden states



$$c_t = \sum_{i=1}^{N} w_i(t) h_i$$

- This solution will work if the weights w_{ki} can somehow be made to "focus" on the right input word
  - E.g., when predicting the word "apfel", $w_3(4)$, the weight for "apple" must be high while the rest must be low
- How do we generate such weights??

Introduction to Attention Mechanism

Introduction to Attention Mechanism

Core Idea: Let the decoder **focus on different parts of the input** sequence at each step of decoding.

- At each decoding step, compute a **weighted sum** over all encoder hidden states.
- Weights reflect **relevance** of each input word to the current output word.

"Soft search" over inputs → more context-awareness.

Mathematical Formulation of Attention

1. **Alignment Score:**

$$e_{t,s} = \text{score}(h_t^{\text{dec}}, h_s^{\text{enc}})$$

2. **Attention Weights (Softmax):**

$$\alpha_{t,s} = \frac{\exp(e_{t,s})}{\sum_{s'} \exp(e_{t,s'})}$$

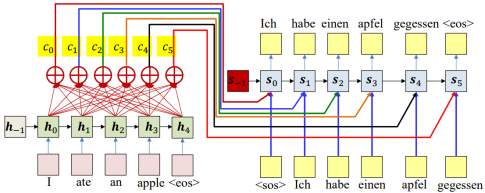
3. **Context Vector:**

$$c_t = \sum_s \alpha_{t,s} h_s^{\text{enc}}$$

4. **Decoder Input:**

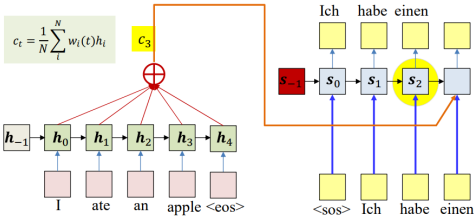
$$y_t = \text{Decoder}(y_{t-1}, h_{t-1}^{\text{dec}}, c_t)$$

Attention Models



- **Attention weights:** The weights are dynamically computed as functions of decoder state.
- **Expectation:** if the model is well-trained, this will automatically "highlight" the correct input.
- But how are these computed?

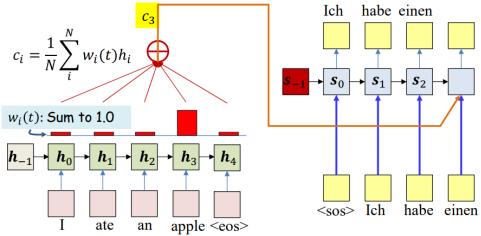
Attention weights at time t



- The "attention" weights $w_i(t)$ must be computed from available information at time t.
- The primary information is s_{t-1} (the state at time $t-1$).
- Also, the input word at time t, but generally not used for simplicity.

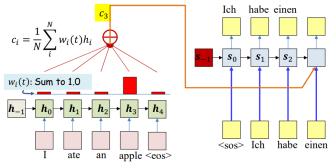
$$c_t = \sum_{i=1}^{N} w_i(t) h_i$$

Requirement on attention weights



- The weights must be positive and sum to 1.0 (i.e., be a distribution).
- Ideally, they must be high for the most relevant inputs for the *i*-th output and low elsewhere.
- **Solution: A two step weight computation**
  - First compute raw weights (which could be +ve or –ve).
  - Then softmax them to convert them to a distribution.

Requirement on attention weights



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  - Ideally, they must be high for the most relevant inputs for the ith output and low elsewhere
- Solution: A two step weight computation
  - First compute *raw* weights (which could be +ve or –ve)
  - Then softmax them to convert them to a distribution

$$e_i(t) = g(\boldsymbol{h}_i, \boldsymbol{s}_{t-1})$$

$$w_i(t) = \frac{\exp(e_i(t))}{\sum_j \exp(e_j(t))}$$

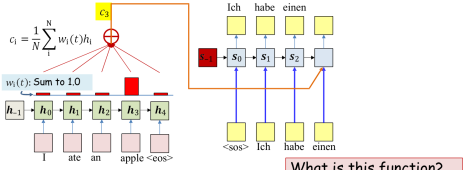
Quiz: Attention Framework

1. The attention framework computes a different "context" vector at each output step?
2. The context vector is chosen as the hidden representation of the word with the highest weight?
3. The weight is a function of the encoder representation and the most recent decoder state?

Quiz: Answer

1. The attention framework computes a different "context" vector at each output step? **True**
2. The context vector is chosen as the hidden representation of the word with the highest weight? **False**
3. The weight is a function of the encoder representation and the most recent decoder state? **True**

Requirement on attention weights



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- Solution: A two step weight computation
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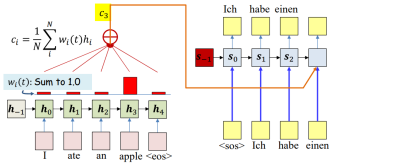
What is this function?

$$e_i(t) = g(\boldsymbol{h}_i, s_{t-1})$$

$$w_i(t) = \frac{\exp(e_i(t))}{\sum_j \exp(e_j(t))}$$

Attention weights



- Typical options for $g()$ (**variables in red must learned**)

$g(\boldsymbol{h}_i, \boldsymbol{s}_{t-1}) = \boldsymbol{h}_i^T \boldsymbol{s}_{t-1}$

$g(\boldsymbol{h}_i, \boldsymbol{s}_{t-1}) = \boldsymbol{h}_i^T \boldsymbol{W}_g \boldsymbol{s}_{t-1}$

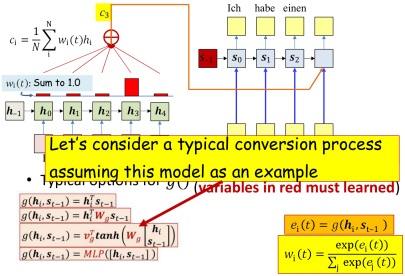
$g(\boldsymbol{h}_i, \boldsymbol{s}_{t-1}) = \boldsymbol{v}_g^T \boldsymbol{tanh}\left(\boldsymbol{W}_g \begin{bmatrix} \boldsymbol{h}_i \\ \boldsymbol{s}_{t-1} \end{bmatrix}\right)$

$g(\boldsymbol{h}_i, \boldsymbol{s}_{t-1}) = \boldsymbol{MLP}([\boldsymbol{h}_i, \boldsymbol{s}_{t-1}])$

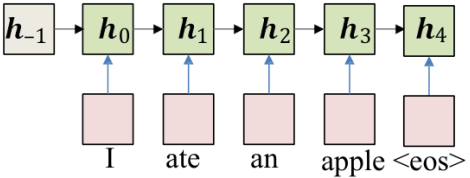
$e_i(t) = g(\boldsymbol{h}_i, \boldsymbol{s}_{t-1})$

$w_i(t) = \frac{\exp(e_i(t))}{\sum_j \exp(e_j(t))}$

Attention weights



Converting an input: Inference



Pass the input through the encoder to produce hidden representations h_i

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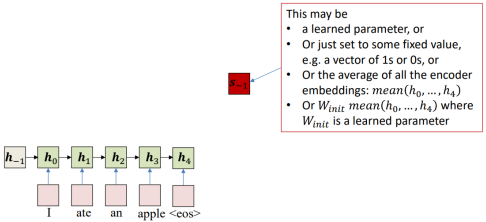

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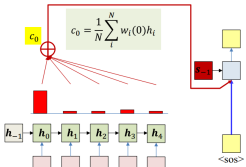
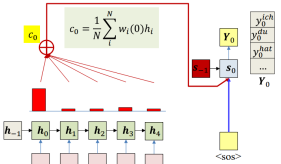


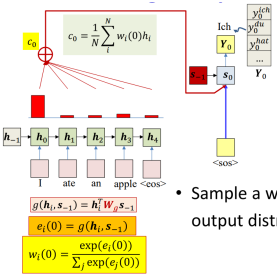
- Compute the attention weights $w_i(0)$ for the first output using s_{-1}
  - Will be a distribution over the input words
- Compute "context" c_0
  - Weighted sum of input word hidden states
- Input c_0 and <sos> to the decoder at time 0
  - <sos> because we are starting a new sequence
  - In practice we will enter the *embedding* of <sos>

Converting an input: Inference



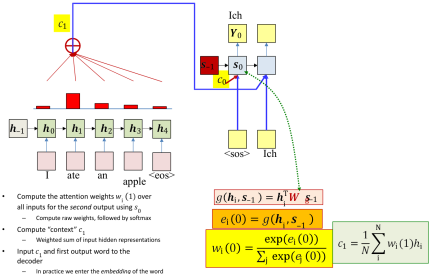
- The decoder computes
  - s_0
  - A probability distribution over the output vocabulary
    - Output of softmax output layer

Converting an input: Inference



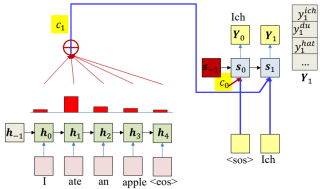
- Sample a word from the output distribution

Converting an input: Inference



- Compute the attention weights $w_i(1)$ over all inputs for the *second* output using s_0
  - Compute raw weights, followed by softmax
- Compute "context" c_1
  - Weighted sum of input hidden representations
- Input c_1 and first output word to the decoder
  - In practice we enter the *embedding* of the word

Converting an input: Inference



- The decoder computes
  - s_1
  - A probability distribution over the output vocabulary

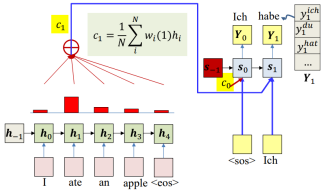
$$g(\boldsymbol{h}_i, \boldsymbol{s}_0) = \boldsymbol{h}_i^T \boldsymbol{W}_g \boldsymbol{s}_0$$

$$e_i(1) = g(\boldsymbol{h}_i, \boldsymbol{s}_0)$$

$$w_i(1) = \frac{\exp(e_i(1))}{\sum_j \exp(e_j(1))}$$

$$c_1 = \frac{1}{N} \sum_i^N w_i(1) h_i$$

Converting an input: Inference



- Sample the second word from the output distribution

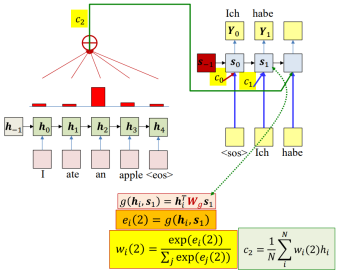
$$g(\boldsymbol{h}_i, \boldsymbol{s}_0) = \boldsymbol{h}_i^T \boldsymbol{W}_g \boldsymbol{s}_0$$

$$e_i(1) = g(\boldsymbol{h}_i, \boldsymbol{s}_0)$$

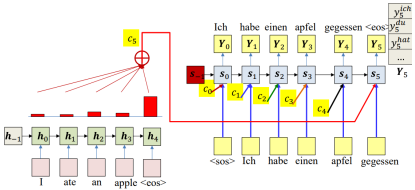
$$w_i(1) = \frac{\exp(e_i(1))}{\sum_j \exp(e_j(1))}$$

$$c_1 = \frac{1}{N}\sum_i^N w_i(1)h_i$$

Converting an input: Inference



Converting an input: Inference



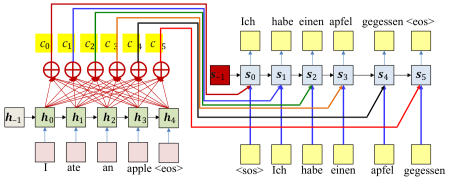
Continue this process until <eos> is drawn

$$e_i(5) = g(h_i, s_4)$$

$$w_i(5) = \frac{\exp(e_i(5))}{\sum_j \exp(e_j(5))}$$

$$c_5 = \frac{1}{N}\sum_i^N w_i(5) h_i$$

Attention-based Decoding

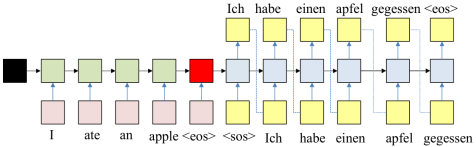


$$e_i(t) = g(\boldsymbol{h}_i, \boldsymbol{s}_{t-1})$$

$$w_i(t) = \frac{\exp(e_i(t))}{\sum_j \exp(e_j(t))}$$

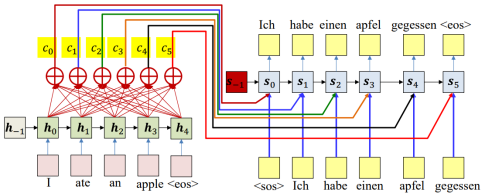
$$c_t = \frac{1}{N}\sum_{i}^{N} w_i(t) h_i$$

Recap: Sequence to Sequence Models



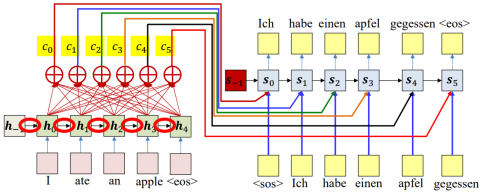
- The input sequence feeds into a recurrent structure
- The input sequence is terminated by an explicit <eos> symbol
  - The hidden activation at the <eos> "stores" all information about the sentence
- Subsequently a second RNN uses the hidden activation as initial state to produce a sequence of outputs

Recap: Attention Models



- Encoder recurrently produces hidden representations of input word sequence
- Decoder recurrently generates output word sequence
  - For each output word the decoder uses a weighted average of the hidden input representations as input "context", along with the recurrent hidden state and the previous output word

Recap: Attention Models



- **Problem:** Because of the recurrence, the hidden representation for any word is also influenced by *all* preceding words
  - The decoder is actually paying attention to the sequence, and not just the word

Impact of Attention

- Improves translation accuracy
- Helps handle **longer sequences**
- Interpretable: shows what the model is focusing on
- Forms the foundation of **Transformer models**

Led to development of:

- Bahdanau Attention (Additive, 2014)
- Luong Attention (Multiplicative, 2015)
- Self-Attention and Transformers (2017+)

Seq2Seq vs Seq2Seq + Attention

Aspect	**Basic Seq2Seq**	**With Attention**
Memory	Single vector (fixed)	Multiple encoder states (dynamic)
Long Sequences	Poor performance	Good performance
Interpretability	Low	High (via attention weights)
Use Cases	Short/medium sequences	Longer, complex tasks

Applications of Seq2Seq + Attention

- Machine Translation
- Text Summarization
- Chatbots and Dialog Systems
- Speech Recognition
- Question Answering
- Video Captioning
- DNA Sequence Modeling

BLEU (Bilingual Evaluation Understudy)

- BLEU is an n-gram overlap metric commonly used to evaluate text generation tasks such as machine translation. It compares generated text against reference text using precision of n-grams.
- Formula:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

where:

- p_n = precision of n-grams
- w_n = weight for each n-gram order (usually uniform)
- BP = brevity penalty to penalize short hypotheses

BLEU (Bilingual Evaluation Understudy) (cont.)

- How it works:
  - BLEU evaluates how many n-grams in the hypothesis match the reference.
  - Missing or misordered words reduce the score.
- Example:
  - Reference: "the cat is on the mat"
  - Hypothesis: "the cat on mat"
  - Unigram precision $= 4/4 = 1.0$
  - Bigram precision $= 1/3 \approx 0.33$
  - BLEU penalizes missing words via BP.
- Why it matters:
  - Easy to compute and widely adopted for automatic evaluation of generated text.

ROUGE (Recall-Oriented Understudy for Gisting Evaluation)

- ROUGE is a recall-focused metric widely used in text summarization to evaluate how much of the reference content is captured by the generated text.
- Common variants:
  - **ROUGE-N:** recall of n-grams.
  - **ROUGE-L:** based on the longest common subsequence.
- Formula (ROUGE-N):

$$\text{ROUGE-N} = \frac{\sum_{\text{n-gram} \in \text{Ref}} \min(\text{count}_{\text{hyp}}, \text{count}_{\text{ref}})}{\sum_{\text{n-gram} \in \text{Ref}} \text{count}_{\text{ref}}}$$

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) (cont.)

- Example:
  - Reference: "the cat sat on the mat"
  - Hypothesis: "the cat sat on mat"
  - ROUGE-1 recall $= 5/6 \approx 83\%$
- Why it matters:
  - Good for checking content coverage in summarization tasks.

METEOR & TER (Other Evaluation Metrics)

- **METEOR:** Measures text generation quality considering not just exact word matches but also stemming, synonyms, and paraphrases. Tends to correlate better with human judgment than BLEU.
- **TER (Translation Edit Rate):** Counts the minimum number of edits (insertions, deletions, substitutions, and shifts) needed to match the reference text. Lower TER indicates better performance.
- Why these metrics matter:
  - Often used alongside BLEU and ROUGE to get a more complete evaluation of generated text quality.

Automated Metrics vs Human Evaluation

- **Automated Metrics** (e.g., BLEU, ROUGE):
  - **Positive:** Cheap, fast, and reproducible.
  - **Negative:** Limited depth; may not capture fluency, naturalness, or true meaning.
- **Human Evaluation:**
  - **Positive:** Captures fluency, naturalness, and semantic meaning.
  - **Negative:** Expensive, slow, and less reproducible.
- Why both matter:
  - Automated metrics allow rapid iteration and benchmarking.
  - Human evaluation ensures quality aligns with real-world perception.
  - Best practice: combine both depending on the task.

Example: BLEU vs ROUGE in Practice

- Reference: “The cat sat on the mat.”
- Hypotheses:
  - Hypothesis A: “The cat is on the mat.”
  - Hypothesis B: “A feline rested on the rug.”
- Metric evaluation:
  - **BLEU:** A ¿ B (closer word overlap with reference)
  - **ROUGE:** A ¿ B (better recall of reference n-grams)
  - Hypothesis B is semantically correct, yet both metrics punish it.
- Takeaway:
  - Different metrics capture different aspects of quality.
  - Word overlap metrics may disagree with human or semantic judgment.

Limitations

- Still sequential — **not fully parallelizable**
- Attention adds computational cost
- Hard to interpret in large-scale models
- Might struggle with very long-range dependencies in huge contexts

Future Directions

- **Transformers**: Fully attention-based, no recurrence
- **Efficient Attention Models**: Longformer, Reformer, Linformer
- **Multimodal Attention**: Vision + Text
- **Memory-Augmented Models**
- **Structured Attention**: Parses, syntax, and alignment bias

Attention paved the way for GPT, BERT, and LLMs

Summary

- Seq2Seq enables mapping input to output sequences of variable lengths
- **Bottleneck:** Fixed-size context vector limits learning capacity
- **Attention:** Improves performance by giving **adaptive access** to input states
- Attention $\rightarrow$ Transformer $\rightarrow$ LLMs
- Still evolving: From additive attention to self-attention and beyond

References

These slides have been adapted from:

- Younes Mourri & Lukasz Kaiser, Natural Language Processing Specialization, DeepLearning.AI
- Bhiksha Raj & Rita Singh, 11-785 Introduction to Deep Learning, CMU

References

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- Sutskever et al. (2014). *Sequence to Sequence Learning with Neural Networks*. NeurIPS.
- Bahdanau et al. (2014). *Neural Machine Translation by Jointly Learning to Align and Translate*.
- Luong et al. (2015). *Effective Approaches to Attention-based Neural Machine Translation*.
- Vaswani et al. (2017). *Attention Is All You Need* (Transformer).
- Chan et al. (2016). *Listen, Attend and Spell* (Speech recognition).

References

Courses & Tutorials:

- Stanford CS224n: Lecture 9 (Attention)
- DeepLearning.ai NLP Specialization (Coursera)
- Harvard NLP Annotated Transformer: http://nlp.seas.harvard.edu/2018/04/03/attention.html
- Jay Alammar's blog: "The Illustrated Transformer"